

Ensor-4_Channels + Raspberry Pi (Kernel 6.12.25)

Preparation Sequence for a Raspberry Pi 5 Board to Work with the "Ensor-4_Channels" Audio Recorder

1. Required Elements

- Raspberry Pi 5
- Original Raspberry Pi Power Supply
- Original 8GB Micro SD Card + USB Micro SD Writer
- 64GB or Larger USB Flash Drive
- A Computer with an Internet Connection and Windows or Linux OS
- The Ensor-4_Channels_RPI5_KERNEL_6.12.25.zip file downloaded from GitHub

2. Install the Raspberry Pi OS Lite (64-bit) operating system without a desktop environment for Raspberry Pi 5

Use Raspberry Pi Imager, configuring the username and password, and

enabling SSH server access

Username: ensor

Password: ***** (any password)

It is recommended to use PuTTY to send commands via SSH to the Raspberry Pi

3. Connect the Ensor-384 board to the Raspberry Pi and then connect the Raspberry Pi to the internet using an Ethernet cable. Power on the system.

3. Update the operating system

```
sudo apt update
```

```
sudo apt full-upgrade
```

4. Run sudo raspi-config

Enable autologin with console in the new kernel (6.12.25) 1 – S6

Enable SSH 3 - I1

Enable I2C 3 - I5

Expand filesystem 6 - A1

```
sudo reboot
```

5. Connect a USB flash drive, with a capacity of 64GB or more, to the Raspi-3b+ board and check its label. The USB flash drive will be formatted later in exFAT.

```
sudo fdisk -l
```

```
> /dev/sda1
```

The USB flash drive will be used to store the audio recordings.

1 hour of audio recording, 2 stereo channels at 192000 m/s and 32 bits, occupies 5.4 GB

1 hour of audio recording, 2 stereo channels at 384000 m/s and 32 bits, occupies 10.8 GB

6. Mount the USB flash drive at system startup

The USB flash drive will be mounted in the following path:

`/media/ensor/Ensor384`

Create the directories `sensor` and `sensor/Ensor384` in `/media`

```
sudo mkdir -p /media/ensor/Ensor384
```

Full permissions enabled for everyone

```
sudo chmod -R 777 /media/ensor/Ensor384
```

Add the following line to the file `/etc/fstab`

```
sudo nano /etc/fstab
```

```
/dev/sda1 /media/ensor/Ensor384 auto auto,user,rw,umask=000,nofail,x-system.device-  
timeout=10 0 0
```

Verify that the modifications in `/etc/fstab` are correct

```
sudo mount -a
```

```
sudo systemctl daemon-reload
```

```
sudo reboot
```

To verify that the USB flash drive mounts correctly

7. The USB flash drive will be formatted in exFAT

This operation can be easily performed in Windows,
or in Linux, as shown below

```
sudo mkfs.exfat -n LABEL /dev/sdXn
```

in our case

```
sudo fdisk -l to see /dev/sdXn
```

```
sudo mkfs.exfat -n Ensor384 /dev/sda1
```

From a Linux computer or Windows:

Download the compressed file `Ensor-4_Channels_RPI5_KERNEL_6.12.25.zip` from the internet.
This file contains the Ensor directory and the audio configuration file `ensor.conf`.

Extract `Ensor-4_Channels_RPI5_KERNEL_6.12.25.zip` and copy the `ensor.conf` file and the Ensor directory to a USB flash drive.

Connect the USB flash drive to the Raspberry Pi and power on the system.

The recording configuration file, `ensor.conf`, will be on the USB flash drive.

Copy the Ensor directory to:

```
/home/ensor/
```

```
cp -dr /media/ensor/Ensor384/Ensor /home/ensor/
```

Once the Ensor directory is copied, you can delete it from the USB flash drive if desired.

```
rm -dr /media/ensor/Ensor384/Ensor
```

For recording files larger than 4GB.

Install the exfat-fuse program on the Raspberry Pi.

```
sudo apt install exfat-fuse
```

8. Copy the compiled DTS file (tlv320adcx140-overlay.dtbo) from the "Ensor-384" audio recording card to /boot/firmware/overlays

The tlv320adcx140-overlay.dts file and the compiled tlv320adcx140-overlay.dtbo file are located in the /home/ensor/Ensor/dts_slave/ directory.

```
/home/ensor/Ensor/dts_slave/tlv320adcx140-overlay.dts
```

You can compile it with the following command:

```
sudo dtc -@ -H epapr -O dtb -o tlv320adcx140-overlay.dtbo -Wno-unit_address_vs_reg  
tlv320adcx140-overlay.dts
```

and copy the compiled file tlv320adcx140-overlay.dtb to /boot/firmware/overlays/

```
sudo cp /home/ensor/Ensor/dts/tlv320adcx140/tlv320adcx140-overlay.dtbo  
/boot/firmware/overlays/
```

9. The Ensor-4_Channel card can also operate in slave mode; therefore, the RPi5 must be in master mode.

For recordings to be successful, you must select the .dts file corresponding to the required sampling rate.

The following .dts files are located in the /home/sensor/Sensor/dts_master folder:

i2s0_4ch_48kHz_32bit_master.dts

i2s0_4ch_96kHz_32bit_master.dts

i2s0_4ch_192kHz_32bit_master.dts

The compiled .dtbo files are also located in the same folder:

i2s0_4ch_48kHz_32bit_master.dtbo

i2s0_4ch_96kHz_32bit_master.dtbo

i2s0_4ch_192kHz_32bit_master.dtbo

For example, to record 4 channels at 192kHz - 32-bit:

Copy the i2s0_4ch_192kHz_32bit_master.dtbo file to /boot/firmware/overlays/

```
sudo cp /home/ensor/Ensor/dts_mastert/ i2s0_4ch_192kHz_32bit_master.dtbo  
boot/firmware/overlays/
```

Modify the /boot/firmware/config.txt file

```
sudo nano /boot/firmware/config.txt
```

```
dtparam=i2c=on
```

```
dtparam=i2s=on
```

```
gpio=4=op,dh
```

```
#Enable only the dtoverlay that is needed
```

```
#dtoverlay=tlv320adcx140-overlay
```

```
#dtoverlay= i2s0_8ch_48kHz_32bit_master
```

```
#dtoverlay= i2s0_8ch_96kHz_32bit_master
```

```
dtoverlay= i2s0_8ch_192kHz_32bit_master
```

```
#Disable the default audio cards
```

```
#dtparam=audio=on
```

```
#The following line cannot be disabled because the screen will not work after restarting
```

```
dtoverlay=vc4-kms-v3d
```

```
#Disable Wi-Fi and Bluetooth to avoid noise and power consumption
```

```
dtoverlay=disable-wifi
```

```
dtoverlay=disable-bt
```

```
=====
```

B.- Restart the system and verify that the ADCX140 recording card is installed

```
arecord -l
```

Verify that card is 0 and device is 0

card 0: Ensor-4_Channels [Ensor-4_Channels], device 0:

C.- Verify that the card records audio at a sampling rate of 192000 m/s and 32-bit

The recording program is located in /home/ensor/Ensor/

```
rpi_master_PIO_Ensor-4_Channels_slave_4ch_32bit.py
```

and the recording configuration file, ensor.conf, is located on the USB flash drive

=====

10. Reboot the system and verify that the ADCX140 recording card is installed

```
arecord -l
```

verify that card is 0 and device is 0

card 0: ADCX140 [ADCX140], device 0:

11. Verify that the card records audio at a sampling rate of 192000 m/s and 32 bits

The recording program is located in /home/ensor/Ensor/

rpi_slave_Ensor-4_Channels_master_2ch_32bit.py

The recording configuration file, `ensor.conf`, is located on the USB flash drive.

If the USB flash drive does not contain the `ensor.conf` configuration file, the `rpi_slave_Ensor-4_Channels_master_2ch_32bit.py` program creates a default one with the data shown below, and the program will run using that data.

By default, the system does not record until "record=yes" is specified.

Recording parameters can be modified with any text editor.

```
nano /media/ensor/Ensor384/ensor.conf
```

[DEFAULT]

record = no #enables recording or not <yes, no>

#by default, does not record

date = 15/5/2024 #date of the recording

```
file_name = test_192kHz_32bits      #filename
```

[sound.format]

sampling = 192000 #sampling frequency in Hz

bits = 32 #number of bits per sample, fixed at 32 bits

```
[record.time]
```

```
total_time = 10           #total recording time in seconds
```

```
file_time = 10 #since the files are very large, it's useful to split them
               into several
```

delay = 0	#wait time, in seconds, from the start, before beginning to record
-----------	--

index = 0 #index of possible recordings

#that are made on the same day, each recording increments

#the index by one

[channel.selection]

#selection of channels to record

channel_1 = yes

#In slave mode, only 2 channels can be selected

channel_2 = no

channel_3 = no

channel_4 = yes

[gain.channels]

#gain of the channels in dB

#the range of allowed values is from 0dB to 42dB

channel_1 = 42

channel_2 = 0

channel_3 = 0

channel_4 = 42

[aux_channels.type]

possible_options = microphone, line_in

aux_ch1_type = microphone

aux_ch2_type = microphone

aux_ch3_type = microphone

aux_ch4_type = microphone

[aux_channels.input_class]

possible_options = differential, single-ended

aux_ch1_input_class = differential

aux_ch2_input_class = differential

aux_ch3_input_class = differential

aux_ch4_input_class = differential

```
[aux_channels.input_impedance]
```

```
possible_options = 2K5, 10K, 20K
```

```
aux_ch1_zin = 2K5
```

```
aux_ch2_zin = 2K5
```

```
aux_ch3_zin = 2K5
```

```
aux_ch4_zin = 2K5
```

A directory structure will be created depending on the current date and the file names will depend on the current hour and minute ending with the file number.

Modify the configuration file to enable recording:

```
nano /media/ensor/Ensor384/ensor.conf
```

```
[DEFAULT]
```

```
record = yes #enables recording or not <yes, no>
```

```
... #by default, it does not record
```

And try recording:

```
sudo /home/ensor/Ensor/rpi_slave_Ensor-4_Channels_master_2ch_32bit.py
```

It is necessary to use sudo to modify the system date.

12. Automatically record at system startup

When the system starts, the eea.py program has a 20-second delay to allow time for the USB flash drive to mount correctly and be ready for use.

In the new kernel, the rc-local.service must be added.

We will use the rc-local.service service.

```
sudo nano /etc/systemd/system/rc-local.service
```

```
[Unit]
```

```
Description=rc-local
```

```
After=network.target
```

```
[Service]
```

```
User=root
```

```
ExecStart=/home/ensor/Ensor/recorder.sh
```

```
[Install]
```

```
WantedBy=multi-user.target
```

Enable the service to start at system startup.

```
sudo systemctl enable rc-local.service
```

13. Script program /home/ensor/Ensor/recorder.sh

```
nano /home/ensor/Ensor/recorder.sh
```

```
#!/bin/sh
```

```
sudo /home/ensor/Ensor/rpi_slave_Ensor-4_Channels_master_2ch_32bit.py
```

```
sudo shutdown -h now
```

This script starts recording according to the sensor.conf configuration and, when finished, shuts down the system to prevent power consumption.

14. Create the tdm_pio service to emulate the TDM protocol

Create the tdm_pio.service file:

```
sudo nano /etc/systemd/system/rc-local.service
```

[Unit]

Description=rc-local

After=network.target

[Service]

User=root

ExecStart=/home/ensor/Ensor/tdm.pio

[Install]

WantedBy=multi-user.target

Enable the service to start automatically at system boot:

```
sudo systemctl enable tdm.pio.service
```

15. Back up the memory card:

Windows: Win32DiskImager

Linux: dd if=/dev/sda of=/path/to/image

From Raspberry Pi: sudo dd if=/dev/mmcblk0 of=/media/ensor/Ensor384/raspbian.img bs=1M status=progress

16. Prepare Raspberry Pi to record at 384,000 samples/s and 32 bits, Rpi5 in slave mode

The 6.12.25 kernel already supports two stereo channels at 384,000 samples per second and 32 bits. Therefore, edit the `sensor.conf` configuration file and add the following:

```
[sound.format]
```

```
sampling = 384000
```

```
bits = 32
```